Formative Usability Evaluation of an Indonesian Home-Services Marketplace Prototype: Task Success, a Ceiling Effect, and a Six-Theme Barrier Taxonomy

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***Abstract*—We report a formative usability evaluation of a high-fidelity, Indonesian-language web prototype for on-demand household maintenance services. Ten participants completed eight standardised tasks across two moderated rounds, yielding eighty task attempts, Single Ease Question (SEQ) ratings on a 7-point scale, and observer commentary. Task success was high at 93.75% (75/80) and mean ease was $M = 6.56$, $SD = 1.05$. We argue that this aggregate is close to uninformative. Computing dispersion per task shows that four of eight tasks have saturated ($SD \leq 0.42$) and can no longer discriminate between a good and a very good design, while one task—locating the FAQ and contact channel—spans the entire scale ($M = 5.70$, $SD = 2.11$, range 1–7) and is also the lowest-scoring. A matched pair of discovery tasks isolates a taxonomy defect from a layout defect: a well-labelled category was found without difficulty ($M = 6.80$, $SD = 0.42$) whereas two near-synonymous labels cost 0.60 points and tripled the variance ($M = 6.20$, $SD = 0.92$). We synthesise six barrier themes with per-corpus denominators and recommend that formative evaluation report per-task dispersion rather than aggregate success. We publish nine threats to validity, four of them severe, and withdraw six unverifiable sources from an earlier draft.**

***Index Terms*—usability testing, Single Ease Question, ceiling effect, formative evaluation, task success, MSME services, Indonesia, think-aloud**

I. Introduction

A. Background and Motivation

Household maintenance services in Indonesia—air-conditioner cleaning, plumbing leak repair, pump servicing, septic tank suction—are dominated by micro and small enterprises whose customer acquisition still runs through *spanduk* (street banners), printed leaflets, and vehicle stickers [6]. That channel has structural properties invisible to both firm and customer: a fixed physical area, capping how much information can be presented; no record of prior work; and no price, which is negotiated privately and is therefore unverifiable in advance. A web interface removes the area constraint—but it does not follow that the resulting interface is good, nor that it is measurable.

Most formative evaluations report a headline success rate, which is a lossy summary of a multidimensional, per-task, per-participant result. This paper shows, on real data, *how much it hides.* Across ten participants and eighty task attempts we find that 93.75% success coexists with (a) four saturated tasks that can no longer generate evidence, (b) one task whose ratings span the entire scale, and (c) a service taxonomy that misleads participants who are otherwise fluent. Our recommendation follows directly: *formative evaluation should report per-task dispersion and defect-probe results, not just totals.* A companion paper [26] treats the price-integrity finding separately, raising a distinct methodological question that this paper sets aside.

B. Research Questions

RQ1. With what ease do young adult users complete a standardised service-discovery and booking sequence? **RQ2.** Which tasks discriminate between participants, and which are saturated? **RQ3.** What are the characteristic barriers, and how frequently does each occur? **RQ4.** Where do participant observation and independent code inspection converge, and where does each uniquely contribute?

C. Related Work

Interface design for Indonesian MSME services. Task-based design and evaluation of e-commerce interfaces is well established in the Indonesian literature [4], [7], [8], [9], [10]; Tianti *et al.* [6] demonstrate a small-business website raising promotional reach for a catering MSME, and Sunarso *et al.* [5] show that visual content raises attraction and conversion in MSME digital marketing. Recent national evidence frames the opportunity: e-commerce platform adoption is associated with MSME income growth [1], while digital management-system adoption [2] and digital literacy [3] predict MSME performance. We adopt the underlying premise—that consistent visual identity and legible service information signal

operational capability—while declining to import effect sizes measured on other stimuli.

Cognitive principles as rationale, not as measured effects. Our design drew on Norman's three levels of action design and the associated grouping and isolation principles [24]. These function as *rationale*: none was manipulated as an independent variable, and we claim no causal effect.

Formative evaluation instruments and their limits. SEQ is a widely adopted post-task instrument—one 7-point item administered immediately after each attempt [25]—recently applied both alone and against SUS [11], [12], [13]. A systematic review of applied usability metrics concluded that instrument choice governs which usability attributes are measurable at all [14]. We adopt SEQ for breadth—one comparable number per attempt—while noting a trade-off our own data exposes sharply (§III-B): SEQ is a *mean* measure, and means conceal exactly what a formative study needs.

Ceiling effects. Item ceiling effects are a documented hazard of bounded self-report scales: they compress variance, bias estimates of change, and can leave a scale unable to separate adjacent levels of quality [15], [16]. We treat this hazard as the object of study rather than a caveat. Relatedly, recent work distinguishes formative studies, which diagnose, from summative studies, which measure against thresholds [17], [18]; this paper is formative throughout, and no variable measuring conversion, revenue, bounce rate, cognitive load, or trust change was instrumented.

Heuristic inspection [19] and its Indonesian applications [20], [21], [22], [23] are complementary but operate on a different object: the code, not the experience. §III-E compares what each method caught.

II. Methods

A. Prototype and Participants

The prototype is a high-fidelity React application localised into Indonesian, implementing service discovery with search and category filtering, service detail pages, a four-stage checkout (summary → schedule → address → payment), a confirmation/invoice view, technician tracking, and an FAQ/contact area. Two properties were left unchanged because they are the subject of the research questions: prices render in **USD** while the interface is Indonesian-language, and service descriptions include **technician-internal procedure content** intended for engineers rather than customers.

Ten participants were recruited and tested in two rounds of five, identified only by anonymous codes (P01–P05, P06–P10). **No source record contains age, gender, occupation, or any other demographic attribute.** The protocol defined participants as Gen Z users aged 18–26, but that criterion **cannot be verified from the data collected**, and one session record reports readability difficulty "for the 40-and-over age range," which is inconsistent with it. We therefore describe the sample as *young adult users recruited purposively* and make **no representativeness claim** for Indonesian Gen Z.

TABLE I
Task completion as recorded and as auditor-corrected.

	Round 1 (n=5)	Round 2 (n=5)	Combined (n=10)
As recorded	39/40 (97.5%)	40/40 (100%)	79/80 (**98.75 %**)
Auditor-corrected	37/40 (92.5%) (headline)	38/40 (95.0%) (headline)	75/80 (**93.75 %**) (headline)

B. Tasks, Measures, and Procedure

Eight tasks span the booking flow (Fig. 1): T1 locate the AC cleaning service; T2 locate the leak-detection service; T3 read the detailed price breakdown; T4 select a booking date and slot; T5 complete the address form; T6 trigger and identify a validation error; T7 navigate back without losing entered data; T8 locate FAQ/contact support. T1 and T2 are *matched* discovery tasks differing only in service category, so their contrast isolates taxonomy from layout; T6 and T7 are *defect probes* with explicit failure criteria. Measures were **task outcome** against a written criterion per task; **SEQ**, 1 = *very difficult* … 7 = *very easy*, immediately after each attempt; and **observer commentary**. Sessions were moderated with concurrent think-aloud. Recording evidently occurred, but **no recording files are present in the repository**; all qualitative material is therefore observer paraphrase, not verbatim transcript.

C. Resolving a Completion-Rate Dispute

Completion was coded during sessions and recorded as 39/40 (round 1) and 40/40 (round 2). Both source recap sheets then appended an auditor note that specific attempts should have been coded failures: in round 1, two T6 attempts recorded as successes although the criterion was *find the validation message* and the notes state the participants did not find it; in round 2, one T6 attempt recorded as success although the note reads "user did not find any error," and another reached confirmation with a street address consisting only of (and). This is an interpretive disagreement about whether failure on a *sub-criterion* constitutes task failure, not an arithmetic error. Rather than adopt either position silently, **we report the auditor-corrected figures as the headline and the as-recorded figures as a sensitivity analysis** (§III-A); the two differ by four attempts—larger than any other coding uncertainty in this study.

III. Results

A. Task Success

All five failures occurred at T6 or T8. A rate above 90% would ordinarily indicate a largely successful design. **We do not read it that way**: the T6 criterion was permissive; success is not integrity, since participants reached confirmation while the interface's own figures were internally inconsistent [26]; and, decisively, success is *ceiling-bound*—four tasks were near-universally completed, so the metric cannot discriminate (§III-B).

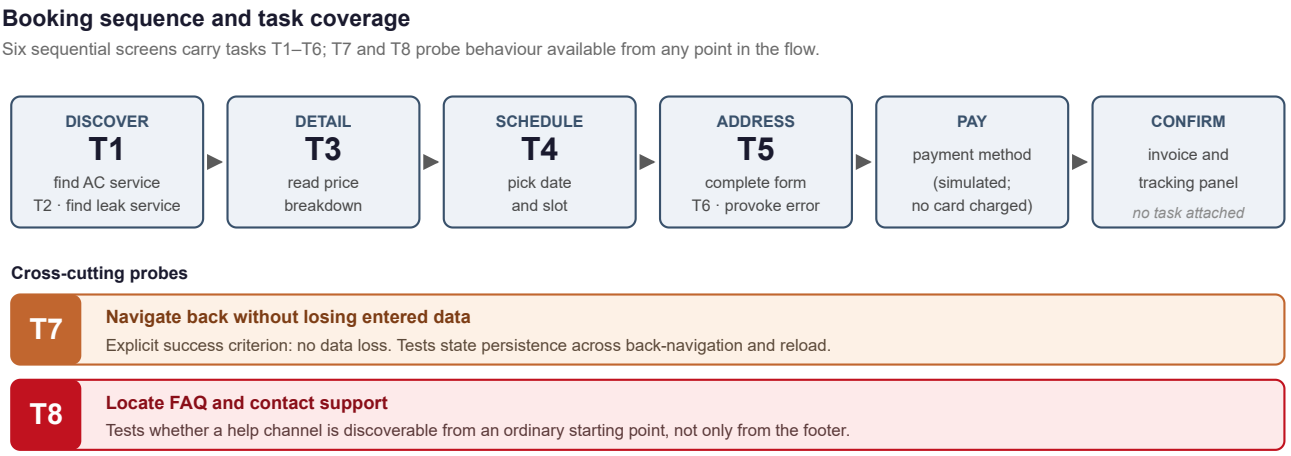


Fig. 1. The eight-task instrument mapped onto the prototype's booking sequence. Six sequential screens carry T1–T6; T7 and T8 probe behaviour available from any point in the flow.

B. Ease, and Why the Mean Is Not the Finding

Overall SEQ: $M = 6.56$, $SD = 1.05$ ($n = 80$), median 7; distribution 62 ratings of 7, 10 of 6, 4 of 5, 1 of 4, 2 of 3, 1 of 1. Two observations follow, and **neither is visible in the aggregate mean.**

(a) Four of eight tasks are saturated. T1, T3, T4, and T5 have $SD \leq 0.42$; for T4 and T5 no participant rated below 6. These tasks cannot separate a good design from a very good one, and further iterations must not rely on them for evidence. Reporting only the mean would have hidden this entirely—and would have invited us to keep iterating on tasks that had already stopped producing signal.

(b) T8 is the study's real signal. Support-channel discoverability has the lowest mean (M = 5.70) *and* by far the widest dispersion (SD = 2.11), spanning the full usable range. The comments locate the cause precisely: participants found the WhatsApp control and read it as a *booking* action rather than a *support* channel; some located help only in the footer; one found no direct contact number at all. Two rated T8 at 7; one rated it 1. **A task whose ratings span the entire scale is under-served by the design, not merely unfamiliar to the participant.** A wide spread invites the reading "some users need training"; the comments refute it. The mechanism is not comprehension but *labelling and placement*—a control that is present, visible, and means something other than it appears to.

C. The Matched Pair Isolates a Taxonomy Defect

T1 and T2 are the same task with a different category. T1 (find AC cleaning) reached $M = 6.80$, $SD = 0.42$; T2 (find leak detection) reached $M = 6.20$, $SD = 0.92$—a 0.60-point drop with triple the variance, on an otherwise identical task. Participants attributed the difficulty to near-synonymous service labels ("smart leak detection" versus pipe inspection) that the interface did not differentiate, and in two cases to the booking control sitting below the fold. This is a **labelling** problem, not a **layout** problem: the category was reachable, but participants could not determine *which* entry addressed their problem. The practical consequence is disproportionate, since merging two catalogue entries is a data change while redesigning the discovery screen is a redesign.

D. Barrier Taxonomy

Six themes emerged. Because the narrative corpus and the SEQ corpus are not confirmed to be the same people (§IV-C, T-1), counts are reported with explicit denominators rather than pooled.

One finding narrows rather than repeats an earlier complaint. On scheduling (T3), a respondent stated that seeing the date *twice* is acceptable if it functions as a cross-check—**provided the two instances agree**. The defect is therefore **disagreement between two date pickers**, not duplication. A single source of truth is consequently a smaller change than removing one picker, and a reviewer who treats duplication as the problem would prescribe the wrong fix.

E. Convergence with Code Inspection

An independent code-based heuristic audit produced 0 Critical, 2 Major, and 4 Minor findings. **Four of six audited defects were independently rediscovered by participants without prompting**, which supports heuristic inspection as a cheap pre-test filter (Table III). The two misses—USD on a localised site, and a total inconsistent with its own line items—motivate the companion paper [26]. Because the audit records these defects as *repaired* while the tested sessions reported all of them, the tested build is probably pre-remediation (§IV-C, T-2).

F. A Contradiction Between Our Own Sources

The narrative report's recommendation matrix rates support/contact remediation **Low** priority; both recap sheets rate it **High**, and four of the narrative's own sessions record difficulty locating support. We adopt the recap sheets' rating, on the

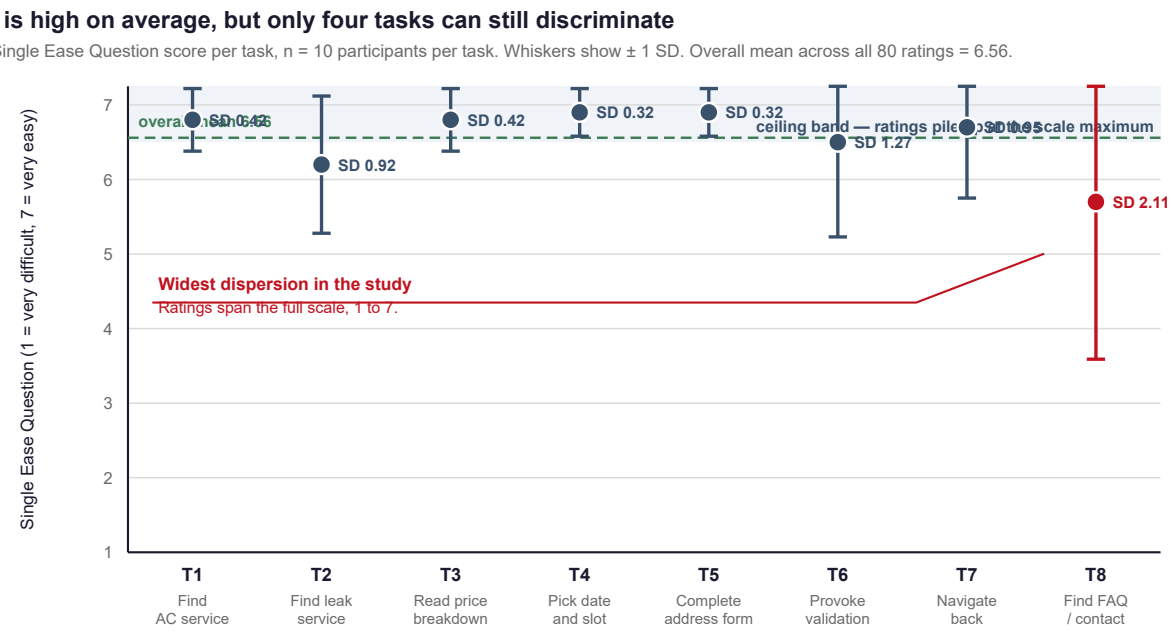


Fig. 2. Mean SEQ per task with ± 1 SD whiskers. Four tasks sit against the scale ceiling with $SD \leq 0.42$. T8 has both the lowest mean and the widest dispersion, spanning the full 1–7 range.

TABLE II
SIX-THEME BARRIER TAXONOMY WITH PER-CORPUS DENOMINATORS. COUNTS ARE NOT POOLED ACROSS CORPORA BECAUSE THE TWO ROSTERS ARE NOT CONFIRMED TO BE THE SAME PARTICIPANTS (§IV-C, T-1).

#	Theme	Narrative (of 10)	SEQ (of 10)	Observed barrier
T1	Currency & technical register	3 named (synthesis asserts unanimity)	2—both round 2, **none in round 1**	USD on an Indonesian site; jargon (“precision diagnostic”, “internal evaporator”, “indoor–outdoor unit relation”) over-excludes lay users; technician-internal documentation shown to customers
T2	Price transparency	3	1 (+1 currency remark)	Price breakdown not collapsible; unit scope (which AC, indoor or outdoor) unstated
T3	Scheduling rigidity	6	5	Date selected twice with **disagreeing** options; 3–5 day window read as weekday-only; no calendar; no arrival range
T4	Address capture & validation	7	5	No map, geocoding, or autocomplete; no saved address; postal code validated but street address not; mandatory fields unmarked; “service address” misread as the provider’s office
T5	Navigation & state persistence	6	8	Form data lost on Back/header navigation and on reload; footer links return to the home page; search query persists after clearing; confirmation button stuck in “processing”
T6a/b/c	Support, confirmation clutter, accessibility	4 / 1 / 2	4 / 1 / 1	Help buried in footer; three-item FAQ; duplicate WhatsApp and print buttons, two tracking bars, unexplained QR code; type size and grey-on-white contrast below threshold

grounds that it is supported by the largest weight of recorded evidence and by the dispersion analysis in §III-B(b). We report the disagreement rather than suppress it, so that a reader checking our recommendation against the primary documents can see that we noticed.

IV. Discussion

A. What Worked

The prototype’s *structure* is sound. Four tasks sit at or near ceiling with almost no dispersion, the booking flow follows an expected mental model—browse, detail, schedule, address, pay, confirm—and participants navigated it without instruction. Note the asymmetry in T5: the address form reached ceiling (M = 6.90, SD = 0.32) **while generating the most frequently requested features in the entire study**—geocoding, autocomplete, saved addresses. The form is easy to fill and wrong in design. **Ease of completion and fitness for purpose are independent properties, and SEQ measures only the first.**

Three patterns organise the failures. **Pattern 1—**

TABLE III
CONVERGENCE BETWEEN CODE-AUDIT FINDINGS AND PARTICIPANT-OBSERVED BARRIERS.

Verdict	Code-audit finding → observed barrier
Converges	Major: home-page price differs from catalogue/detail → T2 unexplained total jumps
Not testable	Major: no visible keyboard focus indicator → —
Converges	Minor: header section links fragile → T5 footer links returned users to home
Converges	Minor: stepper icons contradict text → T6b duplicate tracking status bars
Converges	Minor: no re-order path → T5 participants requested order history
Partly	Minor: postal code silently truncated → T4 postal code validated, street address not
Missed	— → T1 USD currency on a localised site
Missed	— → T2 total ≠ sum of its own line items

TABLE IV
PRIORITISED REMEDIATION BACKLOG. EACH ROW GIVES THE FINDING AND ITS RECOMMENDED ACTION.

P	Finding → recommended action	Sev
1	Total inconsistent with its own line items → add a build-time invariant asserting `total == service + fee + tax`	**Critical**
2	USD on an Indonesian site; technician jargon → localise to IDR; rewrite the taxonomy in lay register; merge near-synonymous entries	**High**
3	Support channel undiscoverable → persistent, explicitly labelled support action in the header; direct contact; FAQ search	**High**
4	Form state lost on Back/reload → persist checkout state; warn before destructive navigation; offer resume	**High**
5	Street address unvalidated; required fields unmarked → validate street address as postal code is; mark required fields; relabel "service address"	**High**
6	Two date pickers that disagree → one source of truth for available dates, with an identical read-only cross-check	**High**
7	Footer links return to home; query persists → fix footer routing; reset the query on clear	**Medium**
8	Confirmation-page clutter → collapse to a static summary; move tracking to order history; label or remove the QR code	**Medium**
9	Type size and contrast below threshold → minimum 12 pt; secondary text to WCAG AA; verify rendered contrast	**Medium**
10	No geocoding, saved address, order history → map pin and autocomplete; addresses stored by label; order history with re-order	**Medium**
11	No path for a missing service; rigid scheduling → "other service" request; full calendar; arrival-time range	**Low**

presentation locale not matched to content locale: an Indonesian-language interface renders prices in dollars and service descriptions in technician-level English. No individual string is wrong; the locale as a whole is incoherent, and participants read it as evidence the product was not built for them. **Pattern 2—missing infrastructure treated as an acceptable default:** no map, no autocomplete, no saved address, no order history, no calendar—collectively, a checkout that asks the user to do work the platform could do, and to do it again next booking. **Pattern 3—redundant, conflicting surfaces:** two date pickers that disagree, a home-page price differing from the catalogue, two tracking bars and an unexplained QR code. Redundancy is normally benign; here it is actively harmful, because when two copies of a fact disagree the user cannot tell which to trust—and in a pricing context, they will trust neither.

B. Methodological Recommendations

- **Report per-task dispersion, not just aggregate success.** A 93.75% success rate coexisted here with a Critical pricing defect and a task rated as low as 1/7.
- **Separate discriminating tasks from saturated ones.** Four of our eight tasks could no longer produce evidence; iterating against them consumes budget for no return.
- **Prefer matched task pairs to isolate causes.** The T1/T2 contrast converted a vague complaint ("search is hard") into a specific, cheap-to-fix diagnosis.
- **Treat effortful completion as a smell.** The address form was easy to fill and wrong. Comfort is not fitness.

C. Threats to Validity

T-1—Participant rosters disagree across source records. [Severe] The narrative corpus names ten participants; the two recap sheets name a partly different set—four overlap, three appear only in the recap sheets, six narrative participants have no SEQ record anywhere. **Consequence:** the corpora cannot be confirmed to be the same ten people. We never aggregate across them as though they were, and every count in §III-D carries an explicit denominator.

T-2—No build identifier, date, or moderator record. [Severe] The protocol required recording the canonical prototype and its build hash before sessions began; no session record contains one, and the tested build probably predates the recorded audit remediation. **Consequence:** no result may be attributed to the current prototype.

T-3—Demographics unrecorded; Gen Z eligibility unverifiable. [Severe] Consequence: for a paper framed around young users, the sample's composition is unestablished.

T-4—Ethics and consent unevidenced. [Severe] No consent record, approval reference, or clearance number appears in any source report. **Consequence:** an unresolved compliance gap that must be closed before publication of this data.

T-5—Ceiling effect limits four tasks. [Moderate] T1, T3, T4, T5 ($SD \leq 0.42$) cannot discriminate between design states; later iterations will need harder tasks or a more complex stimulus.

T-6—Variance collapse between rounds. [Moderate] Round-2 per-task SD (0.49) is $2.8\times$ smaller than round-1 (1.39), with three participants returning near-uniform maximum scores. Differing scoring discipline, moderator effects, demand characteristics, and transcription convention cannot be distinguished from the available records, so we draw **no** design-effect conclusion.

T-7—Qualitative corpus is observer paraphrase, not transcript. [Moderate] Consequence: no direct quotation is used anywhere in this paper.

T-8—Transcription gaps. [Low] One recap sheet notes that a participant's comments were truncated in a screenshot; the check against the original sheet was not performed.

T-9—References partly unverified. [Low] Two citations remain only partially verified (see the note after the reference list); six sources from an earlier draft were withdrawn.

V. Conclusion

We evaluated an Indonesian-language home-services marketplace prototype with ten participants across eight tasks and two rounds. Task success was 93.75% and mean ease 6.56/7—and the most useful result is that **neither number characterises the design.** Beneath the aggregate sit four saturated tasks that can no longer generate evidence, one task spanning the entire scale because a help channel is present but mislabelled, and a matched pair isolating a taxonomy defect from a layout defect. The failures are concentrated and diagnosable: a locale that is Indonesian in language and American in currency, a checkout that discards state on navigation, and two date pickers that disagree with each other.

The lesson generalises beyond this prototype: a formative study reporting only aggregate success rates will systematically under-report exactly the problems that matter most, because discriminating tasks are outnumbered by saturated ones. We recommend publishing per-task dispersion as a matter of course, and we have published ours—with the nine threats to validity that constrain them.

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TABLE V
Complete Single Ease Question matrix.

P	T1	T2	T3	T4	T5	T6	T7	T8	Mean
P01	6	5	7	7	7	7	4	6	6.13
P02	7	7	7	7	7	7	7	7	7.00
P03	7	5	7	7	7	7	7	3	6.25
P04	7	7	7	7	7	7	7	7	7.00
P05	7	5	7	7	7	3	7	1	5.50
P06	6	6	6	6	6	7	7	5	6.13
P07	7	6	7	7	7	7	7	7	6.88
P08	7	7	7	7	7	7	7	7	7.00
P09	7	7	7	7	7	7	7	7	7.00
P10	7	7	6	7	7	6	7	7	6.75
M	**6.80**	**6.20**	**6.80**	**6.90**	**6.90**	**6.50**	**6.70**	**5.70**	**6.56**
SD	0.42	0.92	0.42	0.32	0.32	1.27	0.95	2.11	1.05

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Reference verification. Every reference above was resolved against the Crossref REST API on 2026-10-10; authors, title, journal, volume, issue, pagination and DOI are transcribed from the returned record. Two remain only partially verified and must be checked by hand: [20] (irregular deposited author forms; publisher page range disagrees with earlier drafts) and [21] (author field deposited as malformed "Et. al.").

Appendix A
Instrument, SEQ Matrix, and Provenance

Instrument. A single item, *"After completing the task, how easy or difficult was that task for you?"*, anchored 1 = very difficult and 7 = very easy, administered immediately after each of T1–T8. Ordinal scale; mean and SD are reported for comparability, with the caveat in §IV-C(T-5). Table V reproduces the matrix exactly as recorded in the source recap sheets; recomputation confirmed every reported mean.

Provenance. The full traceability record is maintained in the study's integration record. Six sources cited in the earlier proposal could not be located in Crossref, DataCite, the DOI Handle system, Google Scholar, GARUDA, or SINTA, though several used plausible real author names and real journal titles; all have been removed from the numbered list and the body text, with dependent claims withdrawn, and the one apparently fabricated entry (Hart & Kapp, *J. Des. Res. Pract.*, 2021) has no trace of the article, the authors, or the journal anywhere. **Do not re-add them without retrieving the actual record. Barred figures:** the superseded draft's 87.5%, 4.20/5, 84.6%, and 4.23/5 were never derivable from any source; 84.6% is not the corrected completion rate, and 87.5% appears nowhere in §III-A.